

Malware Fundamentals: A Comprehensive Guide

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DISCLAIMER: Yeh notes sirf educational maqsad ke liye hain. Malware ka gair-kanooni istemal jurm hai.

1. Introduction

Malware, yaani "Malicious Software," digital duniya ka sab se khatarnak pehlu hai. Yeh software ki ek aisi qisam hai jise khas taur par kisi computer, server, ya network ko nuksan pahunchane, data chori karne, ya unauthorized access hasil karne ke liye design kiya gaya hai. Cybersecurity ke student ke liye yeh samajhna zaroori hai ke malware sirf virus nahi hota, balki iski bohat si types hain jo alag-alag tareeqon se kaam karti hain. Malware ka maqsad aksar financial gain, industrial espionage, ya sirf system ko tabah karna hota hai.

2. Malware Lifecycle

Malware ka kaam karne ka ek specific lifecycle hota hai: 1. Infection (system mein ghusna), 2. Execution (code run hona), 3. Persistence (computer restart hone par bhi active rehna), aur 4. Payload Delivery (asli maqsad pura karna jaise data encrypt karna ya chori karna).

3. Types of Malware

Virus (files infect karta hai), Worm (network par khud ko copy karta hai), Trojan (fake software ban kar aata hai), Ransomware (data lock kar ke paise mangna), Spyware (jaasoosi karna), aur Rootkits (system access control karna).

4. Analysis Techniques

Static Analysis (file ko execute kiye baghair code dekhna) aur Dynamic Analysis (Sandbox/Virtual Machine mein malware ko run kar ke uski harkatein check karna).

5. Defense & Protection

Principle of Least Privilege, software updates, EDR tools ka istemal, unknown links/ attachments se door rehna, aur regular backups hi malware se bachne ke behtareen tareeqe hain.

6. Student FAQs

1. Kya antivirus har malware rok sakta hai?

A: Nahi, naye (Zero-day) malware ko pakadne mein antivirus fail ho sakte hain.

2. Kya main malware test kar sakta hoon?

A: Bilkul nahi, sirf dedicated isolated lab mein hi test karein.

3. Malware ki nishani kya hai?

A: PC ka achanak slow hona aur weird pop-ups.

4. Ransomware se bachne ka best tareeqa?

A: Offline backups aur regular system updates.

5. Analysis ke liye tools kaunse hain?

A: Wireshark, ProcMon, aur Ghidra.

6. Kya smartphone safe hain?

A: Nahi, smartphones par bhi malware attack ho sakta hai.

7. Worm aur Virus mein kya farq hai?

A: Virus host file mangta hai, Worm independent hai.

8. Kya Incognito mode safe hai?

A: Nahi, yeh sirf history save nahi karta.

9. Malware developer kaise bante hain?

A: Iska maqsad ethical hacking seekhna hona chahiye, na ke nuksan pahunchana.

10. Phishing kya hai?

A: Phishing ek technique hai jiske zariye malware deliver kiya jata hai.